space-based capsule.[fn4] But that reprieve is short-lived. Before long, the electromagnetic forces that hold together your atoms and molecules cannot hold up against the ever-expanding space within all matter. In the last tiny fraction of a second, molecules crack open, and any thinking beings still holding on are destroyed, torn atom-from-atom from within.

Beyond that point, there is no possibility of watching the destruction, but it carries on nonetheless. Nuclei themselves, the ultra-dense matter in the centers of atoms, are the next to go. The impossibly dense cores of black holes are eviscerated. And at the final instant, the fabric of space itself is ripped apart.

Unfortunately, we may never be able to say with certainty that we are safe from a Big Rip. The problem is that the difference between a Heat Death–fated universe and one headed for a Big Rip might literally be unmeasurable. If dark energy is a cosmological constant, the equation of state parameter w equals -1 exactly, and we get a Heat Death. If w is *at all* lower than -1, even one part in a billion billions, dark energy is phantom dark energy, capable of tearing the universe apart. Because it's impossible to ever measure anything with complete, uncertainty-free precision, the best we may ever be able to do is say that if the Big Rip does occur, it will be so far in the future that all structure in the cosmos will have decayed already by the time it happens. Because even with phantom dark energy, the closer w gets to -1, the farther into the future the Big Rip is pushed. The last time I calculated the earliest possible Big Rip, based on the 2018 data release from the Planck satellite, I got something in the vicinity of 200 billion years.